CAT workshop - Number System

Question: What is the minimum value of n (n 0), such that n! is divisible by (n + 1)?

1) 1 2) 3 3) 5 4) 6

Question: a and b are two distinct real numbers such that LCM(a, b) + HCF(a, b) = ab. How many pairs (a, b) satisfy the given condition?

A.0 B. 1 C. 2 D. 3

Question: If X381 is divisible by 11 and 381Y is divisible by 9, find the smallest values of X and Y.

A. 6, 5 B. 7, 6 C. 6, 8 D. 7, 5

Question: If the sum of any two positive quantities is constant, then their product is maximum when the two quantities

A. are reciprocals of each other B. are a number and its root

C. are equal to each other D. are not equal to each other

Question: Find the greatest number that will divide 400, 435 and 541 leaving 9, 10 and 14 as remainders respectively.

1. 13 2. 17 3. 25 4. 20

Question: How many distinct pairs of 2-digit numbers are possible which have their HCF as 15?

(a) 15 (b) 18 (c) 13 (d) 11

Question: If the LCM of first 20 natural numbers is x, then find the LCM of first 25 natural numbers.

1. 160 x 2. 118 x 3. 115 x 4. 120 x

Question: By which of the following numbers must 34300 be multiplied in order to make it a perfect square?

A. 7 B. 5 C. 6 D. 14

Question: How many zeros are there at the end of the product 33 × 175 × 180 × 12 × 44 × 80 × 66?

A. 2 B. 4 C. 5 D. 6

Question: Suppose, you have 108 green marbles and 144 red marbles. You decide to separate them into packages of equal number of marbles of same colour. Find the maximum possible number of marbles in each package.

A. 4 B. 36 C. 9 D. 24

Question: There are three watches, out of which one loses 4 seconds every minute and one gains 4 seconds every minute. If all the watches are set at 8 a.m. and an alarm is activated, each watch beeps for a second when the time turns 1 minute in that watch. At what time will they beep together again?

A. 9:52 a.m B. 9:48 a.m C. 9:58 a.m D. 10:02 a.m

Question: If the sum of two numbers is 55, and their HCF and LCM respectively are 5 and 120, then the sum of the reciprocals of the numbers is equal to

A. 55/601 B. 601/55 C. 11/210 D. 120/11

Question: Find the remainder when 7^{63} is divided by 342.

A. 0 B. 1 C. 2 D. 3

Question: What is the right most non-zero digit of $13780000^{13780000}$

A. 8 B. 4 C. 2 D. 6

Question: If the number 23x4534x01 is divisible by 11, then how many values can x be substituted with?

A. 1 B. 4 C. 9 D. 10

Question: Find the last digit of the following sum.

1! + 2! + 3! + 4! + ... + 100!

A. 0 B. 3 C. 5 D. 7

Question: What is the last digit of the number obtained by dividing 6^{63} by 4^{23}?

A. 2 B. 4 C. 6 D. 9

Question: If the last two digits of a four-digit number are interchanged, the new number obtained is greater than the original number by 54. What is the difference between the last two digits of the number?

A. 9 B. 12 C. 6 D. Data Inadequate

Question: The product of two numbers is 12,960 and their HCF is 36. How many pairs of such numbers can be found?

A. 3 B. 4 C. 6 D. 2

Question: Find the least number which is divisible by 12, 32, 42, and 63 and also a perfect square.

A. 20,126 B. 4,032 C. 28,224 D. 1,12,896

Question: Two numbers, whose LCM is thrice their HCF. If the sum of the numbers is 280, what is the difference between the two numbers?

A. 70 B. 140 C. 210 D. 280

Question: The sum of two numbers is 100 and their HCF is 5. Which of the following numbers cannot be their LCM?

A. 95 B. 255 C. 455 D. 625

Question: What is the ratio of the LCM of first 35 natural numbers to the LCM of first 38 natural numbers?

A. 1 : 222 B. 1 : 113 C. 1 : 37 D. 1 : 19

Question: A number consists of 3 consecutive digits, such that the digit in the units place being the greatest of the three. The number formed by reversing the digits exceeds the original number by 22 times the sum of the digit. Find the number.

A. 231 B. 234 C. 233 D. 232

Question: Which of the following options is true?

A. $3\sqrt{3}$ is not a rational number.

B. If 'a' is rational and 'n' is an integer > 1, then an is rational.

C. 7 is not the cube of a rational number.

A. A and B B. A and C C. B and C D. A, B and C

Question: What is the remainder when $1923^{1924^{1925}}$ is divided by 1924?

(a) 1922 (b) 1923 (c) 1 (d) 0

Question: If N = $(11111111)^2$, then what is the sum of the digits of N?

A. 54 B. 62 C. 64 D. 68

Question: A is the smallest integer which when multiplied by 3 gives a number made of 5's only. What is the value of C3, if sum of the digits of A is B and sum of the digits of B is C?

(a) 64 (b) 125 (c) 216 (d) 1331

Question: ab and cd are 2 two-digit natural numbers. Given that 4b + a = 13k1 and 5d - c = 17k2, where k1 and k2 are natural numbers. Find the largest number that will always divide the product of ab and cd.

(a) 144 (b) 169 (c) 221 (d) 256

Question: When $33^{34^{35}}$ is divided by 34, the remainder is a and when $33^{34^{35}}$ is divided by 34, the remainder is b. What is the value of b – a?

(a) 0 (b) 32 (c) 48 (d) 64

Question: If we divide a two-digit number by a number consisting of the same digits written in the reverse order, we get 1 as a quotient and 18 as a remainder. If we add 33 in the given number, we get the sum of the squares of the digits constituting that number. Find the number.

1. 79 2. 63 3. 97 4. 87

Question: The HCF of two numbers is 61 and their product is 44652. What is the minimum possible difference between the two numbers?

A. 171 B. 144 C. 61 D. 54

Question: If the LCM of two whole numbers is twice their HCF, then what is the ratio of the larger number to the smaller number?

(a) 3 : 1 (b) 2 : 1 (c) 3 : 2 (d) 4 : 1

Question: P is the smallest number, which when multiplied by 2 is divisible by 6, 24, 39 and 90. Q is the smallest number, which when divided by 2 is divisible by 49, 28, 21 and 35. What is the sum of P and Q?

(a) 8220 (b) 7815 (c) 8436 (d) 8222

Question: Which of the following numbers can never be the LCM of three numbers a, b, c, if a < b < c and their HCF is `8`?

(a) 16 (b) 32 (c) 48 (d) 72

Question: The LCM of a pair of distinct integers is 28. If their HCF is an even number, then how many such pairs of integers are possible?

(a) 4 (b) 3 (c) 5 (d) None of these

Question: The traffic lights at three different road crossings change after every 48 sec, 72 sec and 108 sec. If they all change simultaneously at 09:55:00 hours, then how many times will all three of them change simultaneously between 09:55:00 and 11:30:00?

(a) 13 (b) 15 (c) 16 (d) 18

Question: Three watches are set on an alarm for 6:00 a.m. The alarms ring for 20, 50 and 60 seconds and then go off to the snooze mode for 2, 5 and 6 minutes, respectively. At what time will the alarms start ringing simultaneously for the second time?

(a) 6:30 a.m (b) 6:42 a.m (c) 6:35 a.m (d) 7:12 a.m

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(a) 6:30 a.m (b) 6:42 a.m (c) 6:35 a.m (d) 7:12 a.m

Question: Mr. Chaco buys some chocolates to distribute among poor children. If there are only 3 children, he can distribute all the chocolates equally. Similarly, if there are 4 or 5 children, then also he can distribute all the chocolates equally. If there are 6 children, how many chocolates will each child get if the total number of chocolates is less than 100?

(a) 4 (b) 8 (c) 6 (d) 10

Question: (i) If X381 is divisible by 11, find the value of the smallest natural number X.

(ii) If 381Y is divisible by 9, find the value of the smallest natural number Y.

(a) 6, 5 (b) 7, 6 (c) 6, 8 (d) 7, 5

Question: A number when divided by 296 leaves the remainder 75. If the same number is divided by 37, then what will be the remainder?

(a) 1 (b) 4 (c) 8 (d) 11

Question: What is the right most non-zero digit of $270^{270} + 130^{130}$?

(a) 1 (b) 3 (c) 7 (d) 9

Question: When 2 different numbers are divided by the same divisor, they leave remainders 11 and 21. When their sum is divided by the same divisor, then the remainder is 4. Find the divisor. ?

(a) 36 (b) 28 (c) 14 (d) 10

Question: If x and y are two co-primes, then which of the following is true?

(a) x + 2, y + 2 are also co-primes

(b) 2x, 2y are also co-primes

(c) X^2, y^2 are also co-primes

(d) x − 2, y − 2 are also co-primes

Question: If a + b = c, where a, b and c are positive single-digit integers, what is the greatest possible value of the product abc?

(a) 126 (b) 162 (c) 180 (d) 250

Question: Let A, B and C be positive integers satisfying A < B < C and A + B + C = X. What is the smallest value of X that does not determine A, B and C uniquely?

(a) 9 (b) 6 (c) 7 (d) 8

Question: Let D be a decimal of the form D = 0.abcdabcdabcd …, where the digits a, b, c and d are integers lying between 0 and 9. At most three of these digits are zero. By what number should D be multiplied so that the result is a natural number?

(a) 999 (b) 9990 (c) 49995 (d) 499995

Question: If abc is a three-digit whole number, such that abc = $a^3 + b^3 + c^3$. [300 < abc < 400], find the sum of a, b and c.

(a) 10 (b) 11 (c) 12 (d) Data Sufficient

Question: Find the smallest value of x such that x! ends with exactly 23 zeros.

(a) 100 (b) 99 (c) 98 (d) No such no. exist

Question: Find the maximum value of 'n' at which the expression $\frac{7809!}{504^n}$ will be an integer.

(a) 1949 (b) 1746 (c) 1299 (d) 1088

Question: A number consists of three consecutive digits in order, the one in units place being the greatest of the three. The number formed by reversing the digits exceeds the original number by 22 times the sum of the digits. Find the number.

(a) 345 (b) 567 (c) 123 (d) 234

Question: If the first 100 natural numbers are written side by side to make a big number, how many digits of this number will be even?

(a) 80 (b) 91 (c) 90 (d) 89

Question: If N = 15 × 30 × 45 × 60 × … × 1500, what will be the number of zeros at the end of N?

(a) 63 (b) 55 (c) 97 (d) 124

Question: A number when divided by a divisor left a remainder of 23. When twice the number was divided by the same divisor, the remainder was 11. Find the divisor.

(a) 12 (b) 34 (c) 35 (d) 24

Question: If the LCM of first 60 natural numbers is x, then find the LCM of first 67 natural numbers.

(a) 67x (b) 6387x (c) 4087x (d) 8174x

Question: There is a set of positive integers called Number Box which contains positive integers which when divided by 2, 3, 4, 5, 6 leave the remainders of 1, 2, 3, 4, 5 respectively. What is the maximum value of square root of the number of numbers that can be in the Number Box such that it does not contain any number with 4 digits?

(a) 4 (b) 8 (c) 5 (d) 3

If $(x + 1)(x - 2)(x + 4)$ is the HCF of the polynomials $f(x) = (x^2 + 2x - 8)(x^2 + 4x + p)$ and $g(x) = (x^2 - x - 2)(x^2 + 3x + q)$, what is the value of $(p + q)$?

Question: Three numbers which are co-prime to each other are such that the product of the first two is 551 and that of the last two is 1073. The sum of the three numbers is

(a) 75 (b) 81 (c) 85 (d) 89

Question: Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is

(a) 4 (b) 5 (c) 6 (d) 8

Question: A person had a number of toys to distribute among children. At first, he tried distributing equally between two children, then among three, four, five and six children, but every time he was left with one toy. When he finally tried distributing equally among seven children, he succeeded. What is the smallest number of toys that he could have?

(a) 419 (b) 301 (c) 105 (d) 77

Question: Three rotating wheels can complete 300, 180 and 120 revolutions per 5 minutes. The point of beginning of rotation on each wheel is distinctly marked. On completion of each revolution, the starting point touches the ground. Two coins are tossed when starting marks of all the wheels touch the ground together. Starting from the beginning, how much time would have passed so that the two coins have been tossed 50 times each?

(a) 5 minutes 10 seconds (b) 4 minutes 5 seconds

(c) 6 minutes 5 seconds (d) 6 minutes

Question: Three Vice Presidents (VP) regularly visit the plant on different days. Due to labour unrest, VP (HR) regularly visits the plant after a gap of 2 days. VP (Operations) regularly visits the plant after a gap of 3 days. VP (Sales) regularly visits the plant after a gap of 5 days. The VPs do not deviate from their individual schedules. CEO of the company meets the VPs when all the three VPs come to the plant together. CEO is one leave from January 5th to January 28th 2012. Last time CEO met the VPs on January 3, 2012. When is the next time CEO will meetall the VPs?

A.February 6, 2012 B.February 9, 2012 C.February 7, 2012 D.February 8, 2012 E.None of the above

Question: Two numbers, 297_B and 792_B, belong to base B number system. If the first number is a factor of the second number then the value of B is:

A. 11 B. 12 C. 15 D. 17. E. 19

Question: The micromanometer in a certain factory can measure the pressure inside the gas chamber from 1 unit to 999999 units. Lately this instrument has not been working properly. The problem with the instrument is that it always skips the digit 5 and moves directly from 4 to 6. What is the actual pressure inside the gas chamber if the micromanometer displays 003016?

A. 2201 B. 2202 C. 2600 D. 2960 E. None of the above options

$10000! = (100!)^K \times P$, where P and K $\in$ I. What can be the maximum value of K?

Question: How many numbered pairs (a, b) less than 10 will have HCF other than 1?

Question: If the first 100 natural numbers are written side by side to make a big number, then how many times in this number will the even digits occur?

Let S be a set of positive integers, such that every element n of S satisfies the conditions:

(i) $1000 < n < 1200$
(ii) Every digit in n is odd.

How many elements of S are divisible by 3?

Question: How many three-digit numbers do not change even if their digits are reversed?

A. 81 B. 90 C. 100 D. 10

Question: What is the smallest natural number that can be expressed as the sum of ten consecutive natural numbers as well as the sum of nine consecutive natural numbers?

A. 145 B. 135 C. 125 D. 115

If 35^{128} is divided by 50, then what is the remainder?

Let n (> 1) be a composite integer such that $\sqrt{n}$ is not an integer. Consider the following statements.

A. n has a perfect integer-valued divisor, which is greater than 1 and less than $\sqrt{n}$.
B. n has a perfect integer-valued divisor, which is greater than $\sqrt{n}$ but less than n.

Which of the following is correct?

A. Both A and B are false B. A is true but B is false.

C. A is false but B is true D. Both A and B are true.

a and b are two digits such that both are greater than 1, $a < b$ and $a + b < ab$. Also, $a + b$, $(a + b)^n$ and $(a + b)^{n+1}$ have the same digit at units place. Again, ab, $(ab)^n$ and $(ab)^{n+1}$ have the same digit at the units place. Find the value of a.

Find the remainder when $(257)^{16} + (70)^9 - (137)^{246}$ is divided by 17.

Question: A positive whole number M, less than 100, is represented in base 2 notation, base 3 notation and base 5 notation. It is found that in all the three cases, the last numeral is 1, while in exactly two out of the three cases, the leading numeral is 1. M is equal to

A. 31 B. 63 C. 75 D. 91

Question: Ram and Shyam take a vacation at their grandparents' house. During the vacation, they do any activity together. They either played tennis in the evening or practiced Yoga in the morning, ensuring that they do not undertake both the activities on any single day. There were some days when they did nothing. Out of the days that they stayed at their grandparents' house, they involved in one of the two activities on 22 days. However, their grandmother while sending an end of vacation report to their parents stated that they did not do anything on 24 mornings and they did nothing on 12 evenings.

How long was their vacation?

A. 36 days B. 14 days C. 29 days D. CBD

Question: Let X be a four-digit number with exactly three consecutive digits being same and is a multiple of 9. How many such X's are possible?

A. 12 B. 16 C. 19 D. 20

Question: If a three digit number 'abc' has 2 factors (where a, b, c are digits), how many factors does the 6-digit number 'abcabc' have?

A. 16 B. 24 C. 18 D. 80